Proving Data Structures Correct: Balanced Search Trees

Computer-Assisted Theorem Proving (7CCSACTP)

Mohammad Abdulaziz

2-3 Trees

```
datatype 'a tree23 = Leaf
  | Node2 ('a tree23) 'a ('a tree23)
  | Node3 ('a tree23) 'a ('a tree23) 'a ('a tree23)
```

©Mohammad Abdulaziz

2-3 Trees

```
datatype 'a tree23 = Leaf
  | Node2 ('a tree23) 'a ('a tree23)
  | Node3 ('a tree23) 'a ('a tree23) 'a ('a tree23)
```

Abbreviations:

$$\langle l,\ a,\ r\rangle \equiv \texttt{Node2 l a r}$$

$$\langle l,\ a,\ m,\ b,\ r\rangle \equiv \texttt{Node3 l a m b r}$$

©Mohammad Abdulaziz

2-3 Trees

datatype `'a tree23 = Leaf`
`  | Node2 ('a tree23) 'a ('a tree23)`
`  | Node3 ('a tree23) 'a ('a tree23) 'a ('a tree23)`

Abbreviations:

$$\langle \texttt{l, a, r} \rangle \equiv \texttt{Node2 l a r}$$
$$\langle \texttt{l, a, m, b, r} \rangle \equiv \texttt{Node3 l a m b r}$$

Ordering invariant `bst23`: in $\langle \texttt{l, a, m, b, r} \rangle$,
elements of `l` $<$ `a` $<$ elements of `m` $<$ `b` $<$ elements of `r`

©Mohammad Abdulaziz

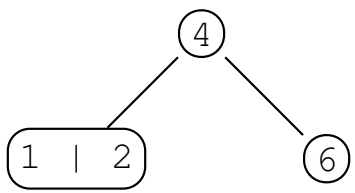
2-3 Trees

datatype `'a tree23 = Leaf`
`| Node2 ('a tree23) 'a ('a tree23)`
`| Node3 ('a tree23) 'a ('a tree23) 'a ('a tree23)`

Abbreviations:

$$\langle \texttt{l, a, r} \rangle \equiv \texttt{Node2 l a r}$$
$$\langle \texttt{l, a, m, b, r} \rangle \equiv \texttt{Node3 l a m b r}$$

Ordering invariant `bst23`: in $\langle$`l, a, m, b, r`$\rangle$,
elements of `l` $<$ `a` $<$ elements of `m` $<$ `b` $<$ elements of `r`



©Mohammad Abdulaziz

Search: `isin`

```
isin Leaf x = False
isin ⟨l, a, r⟩ x = (if x < a then isin l x else
                    if x > a then isin r x
                    else True)
```

Search: `isin`

```
isin Leaf x = False
isin ⟨l, a, r⟩ x = (if x < a then isin l x else
                    if x > a then isin r x
                    else True)

isin ⟨l, a, m, b, r⟩ x =
  (if x < a then isin l x else
   if x = a then True else
   if x < b then isin m x else
   if x = b then True
   else isin r x)
```

©Mohammad Abdulaziz

Search: `isin`

```
isin Leaf x = False
isin ⟨l, a, r⟩ x = (if x < a then isin l x else
                    if x > a then isin r x
                    else True)

isin ⟨l, a, m, b, r⟩ x =
  (if x < a then isin l x else
   if x = a then True else
   if x < b then isin m x else
   if x = b then True
   else isin r x)
```

Assumes the ordering invariant

Structural Invariant `complete`

All leaves are at the same level:

```
complete Leaf = True
complete ⟨l, _, r⟩ = (h(l) = h(r) ∧ complete l ∧ complete r)
complete ⟨l, _, m, _, r⟩ =
  (h(l) = h(m) ∧ h(m) = h(r) ∧
   complete l ∧ complete m ∧ complete r)
```

©Mohammad Abdulaziz

Structural Invariant `complete`

All leaves are at the same level:

```
complete Leaf = True
complete ⟨l, _, r⟩ = (h(l) = h(r) ∧ complete l ∧ complete r)
complete ⟨l, _, m, _, r⟩ =
  (h(l) = h(m) ∧ h(m) = h(r) ∧
   complete l ∧ complete m ∧ complete r)
```

Lemma

`complete t` $\Longrightarrow 2^{h(t)} \leq |t| + 1$

Structural Invariant `complete`

All leaves are at the same level:

```
complete Leaf = True
complete ⟨l, _, r⟩ = (h(l) = h(r) ∧ complete l ∧ complete r)
complete ⟨l, _, m, _, r⟩ =
  (h(l) = h(m) ∧ h(m) = h(r) ∧
   complete l ∧ complete m ∧ complete r)
```

Lemma

`complete t` $\implies 2^{h(t)} \leq |t| + 1$

$\Rightarrow h(t) \leq \log_2(|t| + 1)$: `isin` takes time $O(\log |t|)$